

FileEditSelectionViewGoRun...

AI Lab Programs

AI_Lab_Programs

.dist

.gitattributes

AI_Program_1

AI_Program_1(24BECS124).pdf

AI_Program_2

AI_Program_2(24BECS124).pdf

AI_Program_3

AI_Program_3(24BECS124).pdf

AI_Program_4

AI_Program_4(24BECS124).pdf

AI_Program_5

AI_Program_6

AI_Program_7

AI_Program_8

AI_Program_9

README.md

AI_Program_6

AI_Program_7

AI_Program_8 1, U

AI_Program_9 U

AI_Program_4

AI_Program_9 > NaiveBayes > fit

```
1 class NaiveBayes:
2
3     def __init__(self):
4         self.prior = {}
5         self.conditional = {}
6         self.classes = []
7     def fit(self, data):
8         total_records = len(data)
9         labels = [row[-1] for row in data]
10        self.classes = list(set(labels))
11        # Calculate Priors
12        for c in self.classes:
13            self.prior[c] = labels.count(c) / total_records
14            self.conditional[c] = {}
15        # Calculate Conditionals
16        num_features = len(data[0]) - 1
17
18        for c in self.classes:
19
20            sub_data = [
21                row[:-1]
22                for row in data
```

Problems 1

Output

Debug Console

Terminal

Ports

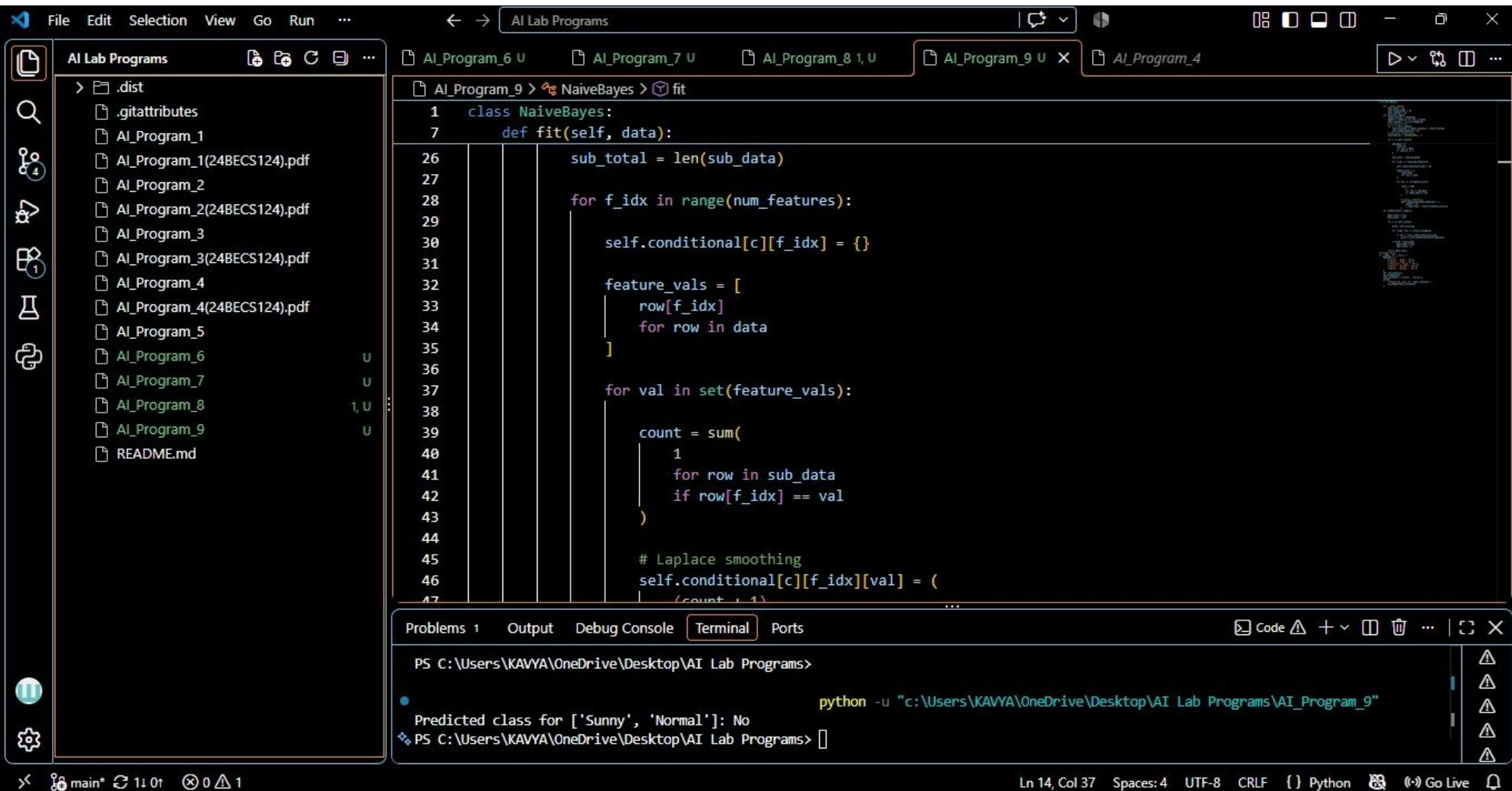
PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs>

python -u "c:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs\AI_Program_9"

Predicted class for ['Sunny', 'Normal']: No

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs>

Ln 14, Col 37 Spaces: 4 UTF-8 CRLF {} Python Go Live



File Edit Selection View Go Run ...

AI Lab Programs

AI Lab Programs

.dist

.gitattributes

AI_Program_1

AI_Program_1(24BECS124).pdf

AI_Program_2

AI_Program_2(24BECS124).pdf

AI_Program_3

AI_Program_3(24BECS124).pdf

AI_Program_4

AI_Program_4(24BECS124).pdf

AI_Program_5

AI_Program_6

AI_Program_7

AI_Program_8

AI_Program_9

README.md

AI_Program_6 U AI_Program_7 U AI_Program_8 1, U AI_Program_9 U AI_Program_4

AI_Program_9 > NaiveBayes > fit

```
1 class NaiveBayes:
50 def predict(self, sample):
52     best_class = None
53     best_prob = -1.0
54
55     for c in self.classes:
56
57         prob = self.prior[c]
58
59         for f_idx, val in enumerate(sample):
60
61             if val in self.conditional[c][f_idx]:
62                 prob *= self.conditional[c][f_idx][val]
63
64             if prob > best_prob:
65                 best_prob = prob
66                 best_class = c
67
68     return best_class
69 # Example Usage
70 if __name__ == "__main__":
71     dataset = [
72         ['Sunny', 'High', 'No'],
73         ['Sunny', 'High', 'No']]
```

Problems 1 Output Debug Console Terminal Ports

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs>

python -u "c:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs\AI_Program_9"

Predicted class for ['Sunny', 'Normal']: No

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs>

Ln 14, Col 37 Spaces: 4 UTF-8 CRLF {} Python Go Live

FileEditSelectionViewGoRun...

AI Lab Programs

AI_Lab_Programs

.dist

.gitattributes

AI_Program_1

AI_Program_1(24BECS124).pdf

AI_Program_2

AI_Program_2(24BECS124).pdf

AI_Program_3

AI_Program_3(24BECS124).pdf

AI_Program_4

AI_Program_4(24BECS124).pdf

AI_Program_5

AI_Program_6

AI_Program_7

AI_Program_8

AI_Program_9

README.md

AI_Program_6

AI_Program_7

AI_Program_8

AI_Program_9

AI_Program_4

AI_Program_9 > NaiveBayes > fit

```
1 class NaiveBayes:
50     def predict(self, sample):
65         best_prob = prob
66         best_class = c
67
68         return best_class
69 # Example Usage
70 if __name__ == "__main__":
71     dataset = [
72         ['Sunny', 'High', 'No'],
73         ['Sunny', 'High', 'No'],
74         ['Overcast', 'High', 'Yes'],
75         ['Rainy', 'Normal', 'Yes'],
76         ['Rainy', 'Normal', 'Yes']
77     ]
78     nb = NaiveBayes()
79     nb.fit(dataset)
80     test_instance = ['Sunny', 'Normal']
81     print(
82         f"Predicted class for {test_instance}:",
83         nb.predict(test_instance)
84     )
```

Problems1OutputDebug ConsoleTerminalPorts

Code + -

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs>

python -u "c:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs\AI_Program_9"

Predicted class for ['Sunny', 'Normal']: No

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs>

Ln 14, Col 37 Spaces: 4 UTF-8 CRLF {} Python Go Live